

Sijil — API Specification

Version: v1 **Base URL:** `/api/v1` **Format:** JSON over HTTPS **Auth:** JWT bearer token
(except `/auth/*`) **Companion documents:** `PRD.md`,
`SOFTWARE_ARCHITECTURE_AND_TECHNICAL_DESIGN.md`

1. Conventions

- All request/response bodies are JSON; `Content-Type: application/json` except file uploads (`multipart/form-data`)
- All timestamps are ISO 8601 UTC (`2026-08-13T09:00:00Z`)
- All IDs are UUIDv4 strings, prefixed by resource type for readability in logs (e.g., `ship_...`, `doc_...`, `li_...`)
- Money values are represented as `{ "amount": "1234.50", "currency": "AED" }` — amount as a string to avoid floating-point precision issues
- Pagination: `?page=1&page_size=25` (default `page_size=25`, max 100); paginated responses follow the shape in Section 3
- Authentication: `Authorization: Bearer <access_token>` header on every request except `/auth/register` and `/auth/login`
- Idempotency: any `POST` that creates a resource accepts an optional `Idempotency-Key` header; replaying the same key returns the original result rather than creating a duplicate

2. Standard Error Response

Every error, regardless of endpoint, follows this shape (matches architecture doc Section 16):

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "quantity must be a positive number",
    "details": {
```

```

    "field": "line_items[0].quantity"
  }
}
}

```

HTTP status	code examples	Retryable
400	VALIDATION_ERROR	No — fix request
401	UNAUTHENTICATED	No — re-authenticate
403	FORBIDDEN	No
404	NOT_FOUND	No
409	CONFLICT (e.g., duplicate trade license on register)	No
422	UNPROCESSABLE (e.g., unreadable document)	No — fix input
429	RATE_LIMITED	Yes, after <code>Retry-After</code> header
502/503	DEPENDENCY_UNAVAILABLE (LLM/OCR provider down)	Yes, with backoff
500	INTERNAL_ERROR	Report to support

3. Standard Paginated Response

```

{
  "items": [ /* ... */ ],
  "page": 1,
  "page_size": 25,
  "total_items": 143,
  "total_pages": 6
}

```

4. Auth

POST /auth/register

Creates a new company and its first admin user.

Auth: None

Request:

```
{
  "company_legal_name": "Al Falah Trading LLC",
  "trade_license_number": "DED-123456",
  "account_type": "trading_company",
  "admin_email": "owner@alfalah.ae",
  "admin_password": "*****",
  "admin_full_name": "Ahmed Al Falah"
}
```

account_type is one of trading_company | broker .

Response 201 :

```
{
  "company_id": "cmp_9f2a...",
  "user_id": "usr_4b1c...",
  "access_token": "...",
  "refresh_token_set": true
}
```

Errors: 400 missing/invalid fields; 409 if trade_license_number already registered.

POST /auth/login

Auth: None

Request:

```
{ "email": "owner@alfalah.ae", "password": "*****" }
```

Response 200 :

```
{
  "access_token": "eyJ...",
  "expires_in": 900,
  "user": {
```

```
    "id": "usr_4b1c...",
    "company_id": "cmp_9f2a...",
    "role": "company_admin"
  }
}
```

Refresh token is set as an httpOnly cookie, not returned in the body.

Errors: 401 invalid credentials.

POST /auth/refresh

Auth: Refresh token cookie **Response** 200 : new access_token , expires_in .

POST /auth/logout

Auth: Required Invalidates the current refresh token. **Response** 204 .

5. Companies & Users

GET /companies/me

Returns the authenticated user's own company profile.

Response 200 :

```
{
  "id": "cmp_9f2a...",
  "legal_name": "Al Falah Trading LLC",
  "trade_license_number": "DED-123456",
  "account_type": "trading_company",
  "broker_company_id": null,
  "created_at": "2026-06-01T08:00:00Z"
}
```

GET /companies/{company_id}/users

Lists users within a company. **Auth:** company_admin or broker_admin for a managed company.

Response 200 : paginated list of { id, email, full_name, role, created_at } .

POST /companies/{company_id}/users

Invites a new user to the company. **Auth:** company_admin.

Request: { "email": "...", "full_name": "...", "role": "company_user" } **Response 201 :** the created (pending-invite) user object.

PATCH /companies/{company_id}/users/{user_id}

Updates a user's role or deactivates them. **Auth:** company_admin.

GET /brokers/{broker_company_id}/managed-companies

Lists companies managed by a broker. **Auth:** broker_admin or broker_user (scoped to their assignments).

6. Shipments

POST /shipments

Creates a new shipment record (the container for a document set + all downstream results).

Auth: Required (company_user or above; broker users create on behalf of a managed company via on_behalf_of_company_id)

Request:

```
{
  "direction": "import",
  "on_behalf_of_company_id": null,
  "notes": "Q3 electronics restock from Shenzhen"
}
```

direction is import | export.

Response 201 :

```
{
  "id": "ship_1a2b...",
  "company_id": "cmp_9f2a...",
  "direction": "import",
  "status": "created",
  "created_at": "2026-08-13T09:00:00Z"
}
```

GET /shipments

Lists shipments for the caller's company (or all managed companies, for a broker).

Query params: `status`, `direction`, `page`, `page_size`

Response 200 : paginated list of shipment summaries.

GET /shipments/{shipment_id}

Full shipment detail, including current `status` (per the state machine in the architecture doc, Section 30) and links to sub-resources.

Response 200 :

```
{
  "id": "ship_1a2b...",
  "company_id": "cmp_9f2a...",
  "direction": "import",
  "status": "classifying",
  "documents": [ { "id": "doc_7c3d...", "doc_type": "commercial_invoice", "status": "pending_review", "line_item_count": 4, "discrepancy_count": 1, "created_at": "2026-08-13T09:00:00Z", "updated_at": "2026-08-13T09:04:12Z" } ]
}
```

DELETE /shipments/{shipment_id}

Soft-deletes a shipment (retained per the retention policy in the architecture doc, Section 29, not hard-deleted). **Auth:** `company_admin`.

7. Documents

POST /shipments/{shipment_id}/documents

Uploads a document for OCR/extraction. **Auth:** Required. **Content-Type:** `multipart/form-data`

Form fields:

- `file` — PDF/JPG/PNG, max 20MB

- `doc_type` — one of `commercial_invoice` | `packing_list` | `bill_of_lading` | `air_waybill`

Response 202 :

```
{
  "document_id": "doc_7c3d...",
  "status": "queued",
  "estimated_completion_seconds": 12
}
```

Errors: 400 unsupported file type/size; 422 file unreadable (corrupt, unsupported scan quality) — surfaced immediately, before queueing, per the architecture doc's edge-case handling (Section 31).

GET `/shipments/{shipment_id}/documents/{document_id}`

Response 200 :

```
{
  "id": "doc_7c3d...",
  "doc_type": "commercial_invoice",
  "status": "extracted",
  "extraction_confidence": 0.94,
  "extracted_fields": {
    "invoice_number": "INV-2026-0451",
    "seller": "Shenzhen Tech Co Ltd",
    "buyer": "Al Falah Trading LLC",
    "total_value": { "amount": "18500.00", "currency": "USD" },
    "line_items": [ /* ... */ ]
  }
}
```

`status values:` `queued` → `processing` → `extracted` | `needs_manual_review` | `failed`.

PATCH `/shipments/{shipment_id}/documents/{document_id}`

Allows manual correction of extracted fields when `status = needs_manual_review`. **Auth:** Required.

Request: partial `extracted_fields` object with corrected values. **Response 200 :** updated document, `status` transitions toward `extracted` once corrected.

8. Line Items & Classification

GET /shipments/{shipment_id}/line-items

Response 200 : paginated list of line items derived from the shipment's documents.

```
{
  "items": [
    {
      "id": "li_2b1c...",
      "description": "24-port managed network switch",
      "quantity": "10",
      "unit_value": { "amount": "185.00", "currency": "USD" },
      "classification": {
        "top_candidate": {
          "hs_code": "8517.62.00.90",
          "confidence": 0.91,
          "reasoning": "Classified under 8517.62 per GRI 1 as apparatus for trans
        },
        "alternatives": [
          { "hs_code": "8471.80.00.00", "confidence": 0.34 }
        ],
        "requires_manual_review": false,
        "user_override_hs_code": null
      }
    }
  ]
}
```

GET /shipments/{shipment_id}/line-items/{line_item_id}/classification

Same classification object as above, in isolation — useful for polling a single item.

POST /shipments/{shipment_id}/line-items/{line_item_id}/override

User manually overrides the suggested HS code. **Auth:** Required. Both the AI suggestion and the override are retained (never overwritten) for audit purposes.

Request:

```
{ "hs_code": "8517.62.00.90", "reason": "Confirmed with our usual customs broker"
```

Response 200 : updated classification object with `user_override_hs_code` populated.

POST /shipments/{shipment_id}/line-items/{line_item_id}/reclassify

Forces a re-run of classification (e.g., after correcting the underlying extracted description).

Auth: Required.

Response 202 : { "status": "processing" }

9. Consistency Checks

GET /shipments/{shipment_id}/discrepancies

Response 200 :

```
{
  "items": [
    {
      "id": "disc_5e6f...",
      "field": "quantity",
      "severity": "blocking",
      "documents_involved": ["doc_7c3d...", "doc_8d4e..."],
      "description": "Commercial invoice states 10 units; packing list states 12",
      "suggested_resolution": "Confirm actual shipped quantity with your supplier"
    }
  ]
}
```

`severity` is one of `blocking` | `warning` | `informational` (architecture doc Section 2.3). A shipment cannot move to `analysis_complete` status while any `blocking` discrepancy is unresolved.

POST /shipments/{shipment_id}/discrepancies/{discrepancy_id}/acknowledge

User explicitly acknowledges a discrepancy rather than resolving it (e.g., a legitimately partial shipment). **Auth:** Required. Logged with the acknowledging user's ID and timestamp.

10. Permit Triage

GET /shipments/{shipment_id}/permits

Response 200 :

```
{
  "items": [
    {
      "id": "perm_3a4b...",
      "regulator": "TDRA",
      "permit_type": "Telecom equipment type approval",
      "applies_to_line_items": ["li_2b1c..."],
      "estimated_processing_time_days": 10,
      "reference_link": "https://tdra.gov.ae/..."
    }
  ],
  "no_permits_required": false
}
```

If no category-specific permits apply, `items` is empty and `no_permits_required: true` is returned explicitly — the architecture doc’s Section 2.4 requires a clear “no permits required” state rather than an ambiguous empty list.

11. CEPA Origin Determination

GET `/shipments/{shipment_id}/line-items/{line_item_id}/origin`

Response 200 :

```
{
  "line_item_id": "li_2b1c...",
  "agreement": "UAE-India CEPA",
  "qualifies": true,
  "reasoning": "Regional value content of 42% exceeds the 35% threshold required",
  "required_documents": ["Preferential Certificate of Origin (India)"],
  "estimated_duty_savings": { "amount": "450.00", "currency": "AED" }
}
```

If value-content data wasn’t provided, `qualifies` returns `"insufficient_data"` (not `false`) with a `missing_fields` array — per the architecture doc’s explicit requirement to never default a missing-data case to a false negative (Section 6.4).

POST `/shipments/{shipment_id}/line-items/{line_item_id}/origin/value-breakdown`

Supplies the local/imported value-content breakdown needed to complete an origin determination.

Request:

```
{ "local_content_value": { "amount": "7770.00", "currency": "USD" }, "total_value
```

Response 200 : updated origin determination (triggers recalculation).

12. Reports

POST /shipments/{shipment_id}/report

Triggers generation of the full compliance report (classification + discrepancies + permits + origin, bundled).

Response 202 :

```
{ "report_id": "rep_6f7g...", "status": "generating" }
```

GET /reports/{report_id}

Response 200 (once ready):

```
{
  "id": "rep_6f7g...",
  "shipment_id": "ship_1a2b...",
  "status": "ready",
  "download_url": "https://.../reports/rep_6f7g....pdf",
  "generated_at": "2026-08-13T09:12:00Z"
}
```

status: generating | ready | failed. download_url is a short-lived signed URL, regenerated on each request rather than permanently public.

13. Billing

GET /billing/subscription

Current subscription status for the caller's company.

Response 200 :

```
{
  "plan": "growth",
  "status": "active",
  "shipments_included_per_month": 100,
  "shipments_used_this_period": 34,
  "current_period_end": "2026-09-01T00:00:00Z"
}
```

POST /billing/checkout

Initiates a subscription checkout session with the billing provider. **Auth:** `company_admin`.

Request: { "plan": "growth" } **Response 200 :** { "checkout_url": "https://billing-provider.../session/..." } — frontend redirects the user there; Sijil never collects card details directly (architecture doc Section 15).

POST /billing/webhook

Receives billing-provider webhooks (subscription updated/cancelled/payment failed). **Auth:** Signature-verified, not user-authenticated. Internal use only — not called by frontend clients.

14. Status Streaming (shipment processing updates)

GET /shipments/{shipment_id}/events (Server-Sent Events)

Streams status transitions for a shipment as they happen, so the frontend doesn't need to poll aggressively during document processing.

Response: `text/event-stream`

```
event: status_update
data: {"shipment_id": "ship_1a2b...", "status": "classifying"}

event: status_update
data: {"shipment_id": "ship_1a2b...", "status": "analysis_complete"}
```

SSE is used rather than WebSockets deliberately — this is one-directional server-to-client updates only, and SSE is the simpler mechanism for that (architecture doc Section 10).

15. Rate Limits

Scope	Limit	Notes
Auth endpoints (per IP)	10 requests / minute	Brute-force mitigation
Document upload (per company)	Tunable per plan tier	Controls LLM/OCR cost exposure — exact numbers are an open assumption per the architecture doc, to be tuned against real usage
All other endpoints (per user)	300 requests / minute	Generous default; revisit if abuse patterns emerge

Rate-limited responses return `429` with a `Retry-After` header.

16. Versioning

The API is versioned via the URL path (`/api/v1/...`) from day one, per the architecture doc's Section 10 rationale — this allows introducing `/api/v2/...` for breaking changes later without disrupting existing integrations (e.g., once enterprise customers begin direct API integration, per the architecture doc's Section 32 future-scope notes).

17. Open Items

- Exact per-plan rate limits (Section 15) — pending real LLM cost data
- Whether a webhook system for *outbound* notifications (e.g., notify a customer's own system when a shipment's status changes) is needed — not in MVP scope, but a natural extension once enterprise API access (architecture doc Section 32) is prioritized
- OpenAPI/Swagger schema generation from the FastAPI app — recommended once the routes above stabilize past MVP, so this document and the live schema stay in sync automatically rather than drifting